Problem 1:

(1) $4591 = 3549 + 1042$

$3549 = 1042 \times 3 + 423$

$1042 = 423 \times 2 + 196$

$423 = 196 \times 2 + 31$

$196 = 31 \times 6 + 10$

$31 = 10 \times 3 + 1$

$\Longrightarrow \gcd(4591, 3549) = 1.$

(2) $1 = 31 - 10 \times 3$

$= 31 - 3 \times (196 - 31 \times 6)$

$= 19 \times 31 - 3 \times 196$

$= 19 \times 423 - 41 \times 196$

$= 101 \times 423 - 41 \times 1042$

$= 101 \times 3549 - 344 \times 1042$

$= 445 \times 3549 - 344 \times 4591$

(3) $\forall\, a, b \in \mathbb{Z}$, $\exists\, x, y \in \mathbb{Z}$ such that

$$\gcd(a, b) = ax + by.$$

If n is a common divisor of a & b, then $n \mid a$ and $n \mid b$.

It follows that $n \mid ax + by$. Hence $n \mid \gcd(a, b)$.

Take $a = 4591$ and $b = 3549$.

Problem 2

(1) $136 = 2^3 \times 17$; $150 = 2 \times 3 \times 5^2$; $255 = 3 \times 5 \times 17$;

$713 = 23 \times 31$; $3549 = 3 \times 7 \times 13^2$

(2) $\gcd(136, 150) = 2$, $\operatorname{lcm}(136, 150) = 2^3 \times 3 \times 5^2 \times 17 = 10200$.

$\gcd(255, 3549) = 3$, $\operatorname{lcm}(255, 3549) = 3 \times 5 \times 7 \times 13^2 \times 17 = 301665$.

Problem 3.

(a) Prove by induction on n.

Base case: $n=2$. See slides.

Induction hypothesis: '$n-1$' holds, $n \geq 3$. If $p \mid a_1 a_2 \cdots a_{n-1}$, then $p \mid a_i$ for some i.

Inductive step: $p \mid a_1 \cdots a_n \Rightarrow p \mid (a_1 a_2 \cdots a_{n-1}) a_n$

$\text{(Base case)} \Rightarrow p \mid a_1 a_2 \cdots a_{n-1}$ or $p \mid a_n$

$\text{(IH)} \Rightarrow p \mid a_i$ for some $1 \leq i \leq n-1$ or $p \mid a_n$

$\Rightarrow p \mid a_i$ for some i.

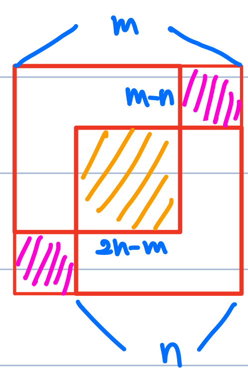
(c) $p \mid n^2 \Rightarrow p \mid n \cdot n \Rightarrow p \mid n$ or $p \mid n \Rightarrow p \mid n$.

(b) If $\sqrt{2}$ is rational, then $\sqrt{2} = m/n$ for some $m, n \in \mathbb{N}_*$, or $2n^2 = m^2$. ①

It follows that $m > n$, $2n > m$, and $2(m-n)^2 = (2n-m)^2$. ②

Consider the equation $2x^2 = y^2$: ① & ② imply that if $(x,y) = (n,m)$ is a solution, then so does $(x,y) = (m-n, 2n-m)$. Hence, this equation has no minimum solution in $\mathbb{N}_*$, which is absurd. (Why? Given a solution $(x,y) = (n,m)$, we can check all candidates in $\{(x,y) \in \mathbb{N}_* : x \leq n, y \leq m\}$.)

Conclusion: $2n^2 = m^2$ has no non-negative integer solution, or $\sqrt{2} \notin \mathbb{Q}$.



$2 \cdot$ =

Problem 4. The following picture explains everything: (IH = inductive hypothesis)

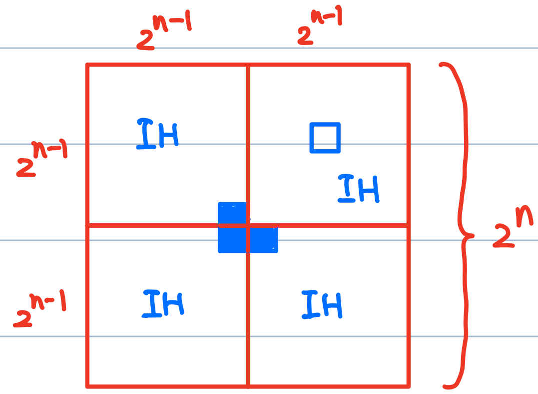


Figure: Inductive step.

Exercise: Find an error in the following proof.

Thm: All horses are of the same color.

Proof: Prove by induction on the number of horses N.

Base case: $N=1$. Trivial since we only have one horse.

Inductive step:

Suppose we have $N+1$ horses and the statement is true for N horses.

Name the horses as

$$H_1, H_2, \cdots, H_N, H_{N+1}.$$

By IH, $\{H_1, H_2, \cdots, H_N\}$ is a set of horses of the same color.

By IH, $\{H_2, \cdots, H_{N+1}\}$ is a set of horses of the same color as well.

Conclusion: all horses have the same color as H_N. #.